ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

AGRICULTURAL MANAGEMENT PRACTICES

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 11 Agricultural Management Practices practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept other correct answers. Give follow-through marks for correct use of an earlier wrong figure.

QUESTION 1

1.1 1.1.1 B ✓✓ 1.1.2 A ✓✓ 1.1.3 A ✓✓ 1.1.4 A ✓✓ 1.1.5 A ✓✓ (10)

1.2 1.2.1 Crossbreeding ✓✓ 1.2.2 Artificial insemination ✓✓ 1.2.3 Field capacity ✓✓ 1.2.4 Value adding ✓✓ 1.2.5 Cooperative ✓✓ (10)

1.3 1.3.1 True ✓✓ 1.3.2 False: calmer / less aggressive ✓✓ 1.3.3 False: Variable ✓✓ 1.3.4 True ✓✓ 1.3.5 False: Crossbreeding ✓✓ (10)

[30]

QUESTION 2

2.1 Mating animals of two different breeds ✓✓. Advantage, any ONE: hybrid vigour (faster growth, better fertility, more resistance to disease) ✓. (3)

2.2 Cow: about 283 days (accept 280–285) ✓; ewe: about 147–150 days ✓; sow: about 114 days (3 months, 3 weeks, 3 days) ✓ (3)

2.3 Days left in March: 30; April 30 (60); May 31 (91); June 30 (121); July 31 (152); August 31 (183); September 30 (213); October 31 (244); November 30 (274) ✓✓; 283 − 274 = 9, so 9 December ✓ (3)

2.4 $\text{Maize: } 44 - 16 = 28 \text{ parts } ✓; \text{ oilcake: } 16 - 9 = 7 \text{ parts } ✓; \text{ total } 35 \text{ parts } ✓$ (6)

$$\text{Maize} = \frac{28}{35} \times 1\,000 = 800 \text{ kg } ✓✓; \text{ oilcake} = \frac{7}{35} \times 1\,000 = 200 \text{ kg } ✓$$

2.5 Any ONE: foot-and-mouth disease / anthrax / bovine tuberculosis / brucellosis / lumpy skin disease ✓. They spread quickly and are very dangerous to animals, people and the economy ✓, so the state must control them (quarantine, movement controls) to stop outbreaks ✓. (3)

2.6 Any THREE: freedom from hunger and thirst ✓; freedom from discomfort ✓; freedom from pain, injury and disease ✓; freedom to express normal behaviour; freedom from fear and distress (3)

2.7 Any TWO: branding ✓; ear tags (permanent registered tags) ✓; tattooing; ear notching; microchips (2)

2.8 To prevent diseases before they occur ✓, reducing deaths, treatment costs and production losses ✓. (2)

[25]

QUESTION 3

3.1 Deciding when to irrigate and how much water to apply ✓✓. Any TWO methods: tensiometers / soil moisture probes ✓; evaporation pan readings ✓; feel method (soil in the hand); watching plants for wilting; weather data (4)

3.2 $\text{Total N} = 120 \times 3 = 360 \text{ kg} \checkmark; \text{LAN} = \frac{360}{0{,}28} \checkmark\checkmark = 1\,285{,}71 \text{ kg} \checkmark\checkmark$ (5)

3.3 IPM combines several methods to keep pests below a harmful level ✓, using chemicals only when needed, to protect the environment ✓. Any THREE: biological control (natural enemies) ✓; cultural control (crop rotation, resistant varieties, clean fields) ✓; mechanical control (traps, hand-picking) ✓; careful, targeted chemical control (5)

3.4 Any THREE: husks are dry and brown ✓; a black layer forms at the base of the kernels (physiological maturity) ✓; kernels are hard and dry, with moisture content about 12–14% ✓; leaves are dry (3)

3.5 Any TWO causes with prevention: insects (weevils) ✓ → clean, sealed storage / fumigation ✓; moulds from moisture ✓ → dry the grain properly before storage ✓; rodents → rat-proof stores; poor handling → careful harvesting and packaging (4)

3.6 Any FOUR: production all year round ✓; control of temperature and humidity ✓; protection from frost, hail and wind ✓; higher yields and quality ✓; fewer pests and diseases; less water lost (4)

[25]

QUESTION 4

4.1 The gross income of an enterprise minus its variable (direct) costs ✓✓. (2)

4.2 (8)

(a) $1\,000 \times 2 \times \text{R}38 \checkmark = \text{R}76\,000 \checkmark$

(b) $9\,000 + 38\,000 + 2\,500 + 3\,000 + 6\,000 \checkmark = \text{R}58\,500 \checkmark$

(c) $76\,000 - 58\,500 \checkmark = \text{R}17\,500 \checkmark$

(d) $17\,500 \div 1\,000 \checkmark = \text{R}17{,}50 \text{ per bird} \checkmark$

4.3 Any TWO, 2 marks each: buy inputs in bulk at lower prices ✓✓; better selling prices and access to bigger markets ✓✓; shared equipment and storage; training and advice; access to credit (4)

4.4 Example: slaughter, clean and package the chickens (or make portions, smoked or marinated products) ✓. Processed products sell at a higher price per kg ✓, so she earns more from each bird ✓. (3)

4.5 Animals Protection Act (Act 71 of 1962) ✓ (accept the Animal Health Act or the Meat Safety Act with a relevant requirement). Requirement, any ONE: provide enough food, water and shelter ✓✓ / no cruelty or unnecessary suffering / humane transport and slaughter. (3)

[20]

TOTAL: 100

END OF MARKING GUIDELINE